Stefano Ribes
Resume

Personal Data

citizenship:	Italian (EU)
email:	ribes.stefano@gmail.com
personal webpage:	https://ribesstefano.github.io/ (Google Scholar, Github, LinkedIn)

Selected Publications

[I] **Ribes, S.**, Dunlop, N., & Mercado, R. (2026). TACK: A Statistical Evaluation of Degradation Activity on a Novel TArgeting Chimeras Knowledge Dataset. In *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2 (KDD'26)*. https://doi.org/10.1145/3770855.3819052, link.

[II] **Ribes, S.**, Zhang, R., Cropsal, T., Källberg, A., Tyrchan, C., Nittinger, E., & Mercado, R. (2026). PROTAC-Splitter: a machine learning framework for automated identification of PROTAC substructures: S. Ribes et al. *Journal of Cheminformatics*, 18(1), 30., link.

[III] **Ribes, S.**, Nittinger, E., Tyrchan, C., & Mercado, R. 2024. Modeling PROTAC degradation activity with machine learning. *Artificial Intelligence in the Life Sciences, 6*, p.100104, link.

Education

2024 - December 2026	PhD Student in Computer Science and Engineering, **AI Laboratory for Molecular Engineering (AIME)**, at Chalmers University, Sweden PI: Rocío Mercado Oropeza | Group page.
2021 - June 2023 Avg. grade: 4.3/5	Master's Degree in **Data Science and AI** at Chalmers University or Technology, Gothenburg, Sweden Thesis: "Machine Learning for Predicting Targeted Protein Degradation"
2016 - 2021	Swedish Licentiate of Technology Degree in **Multi-LSTM Acceleration and CNN Fault Tolerance** at Chalmers University, Sweden Thesis: "Multi-LSTM Acceleration and CNN Fault Tolerance".
2015 - 2016 Avg. grade: 4.25/5	Erasmus Exchange in **Embedded Electronic System Design** at Chalmers University of Technology, Gothenburg, Sweden
2014 - 2016 Final grade: 101/110	Master's Degree in **Computer Engineering, Embedded Systems** at Polytechnic University of Turin, Turin, Italy
2011 - 2014 Final grade: 96/110	Bachelor's Degree in **Computer Engineering** at University of Modena and Reggio Emilia, Modena, Italy Thesis: "*Study and Research on Responsive Technologies for Web Applications*"

Programming Skills and Tools

Strong Experience in:	PyTorch, PyTorch Lightning, Optuna, Pandas, Scikit-learn
High Performance Computing:	C, C++ , CUDA (CuPy and Numba), OpenCL
Version Control:	Proficient in Git for code versioning and collaboration in team projects
Scripting:	I'm confident scripting in both Linux and Windows environments

For more information, please visit my GitHub page: https://github.com/ribesstefano

Work Experience

June 2023 - 2024 **Research Student** at AI Laboratory for Molecular Engineering, Chalmers University of Technology.

At the AIME laboratory, I've continued my master thesis work on predicting protein-ligand activity using deep learning models, mainly language models based on the Hugging Face Transformers library.

Spring 2023 **Master Thesis Intern** at AstraZeneca, Gothenburg, Sweden.

At AstraZeneca, I worked on my thesis titled "Machine Learning for Predicting Targeted Protein Degradation", focusing on designing diverse candidate deep learning models to predict PROTACs targeted protein degradation activity. The thesis can be read at this link.

2020 - 2022 **Hardware Engineer** at Cobham Gaisler, Gothenburg, Sweden.

I was a digital hardware engineer focusing on writing and testing IP modules and machine learning accelerators to be integrated in a RISC-V processor.

Teaching

All courses taught as a teaching assistant at Chalmers University of Technology, Department of Computer Science and Engineering, between 2024 and 2026, primarily through lab sessions and grading.

AI for Molecules Course given by Rocío Mercado. Prepared and coded all programming assignments, delivered weekly 15-minute recitation introducing each assignment, and supported students during assignment hours.

Master's Thesis Supervision Co-supervised three MSc theses on reinforcement learning and generative AI for multi-objective molecular design, and machine-learning-based synthesizability prediction of PROTAC molecules.

Applied Machine Learning Three course instances given by Selpi Selpi. Developed and graded exam questions, in addition to lab sessions.

Introduction to Data Science and AI Given by Simon Olsson and Rocío Mercado.

Statistical Methods for Data Science Given by Yinan Yu.

Design of AI Systems Given by Ashkan Panahi.

Languages

English: Fluent | IELTS 6.5 (2015)
Swedish: Colloquial level
Italian: Mothertongue

Interests and Activities

Good sketching abilities and a fine eye for details,
I love cooking, especially experimenting new cuisines,
I play football as a goalkeeper and I particularly yearn for new challenges.